HARSHAL PRIYAVADAN PATEL

SYSTEMS SAFETY & AI GOVERNANCE RESEARCHER

Human–Machine Accountability · Safety-Constrained Autonomy · Systems Security

Chandler, Arizona

PROFESSIONAL PROFILE

Independent systems researcher, software developer and self-employed technologist working across AI governance, systems safety, security research and financial risk modeling.

Creator of HarshalOS and the Accountability-Gated Systems (AGS++) framework, with research focused on preserving human accountability, constraining autonomous behavior and making system actions traceable. Additional experience includes technical financial analysis using TradingView Pine Script, independent security research, commercial driving, and import/export and wholesale operations.

CORE RESEARCH & TECHNICAL WORK

HarshalOS — Systems Architect & Developer

Self-employed · Independent project

- Designed and implemented a browser-based structural demonstration of accountability-gated systems, including a stateful kernel, transition validation and refusal logic.
- Developed mechanisms for local audit records, hash-chain consistency, evidence handling and structured governance.
- Designed the Human Key demonstration and associated authorization state machine, with explicit limitations on identity verification and production authorization.
- Developed the HOSLLM data-only interface, separating model-generated material from system authority.
- Created security and threat-model documentation addressing browser trust boundaries, authorization, data integrity and deployment risks.
- Distinguished implemented capabilities from simulated components and proposed production requirements.

HarshalOS on GitHub

AGS++ — Original Creator & Independent Researcher

- Created the Accountability-Gated Systems framework and associated structural, communication and governance models.
- Developed a keyed human–machine hybrid systems model focused on identity, authorization, continuity and accountability.
- Formulated principles for bounded autonomy, reversibility, refusal and controlled non-operational states.
- Authored technical documentation, research material, examples and audit-oriented specifications.
- Developed a framework for evaluating accountability and safety properties in AI-assisted and autonomous systems.

INDEPENDENT & PROFESSIONAL EXPERIENCE

Independent Financial Analyst & Risk Modeler

Self-employed · Independent analysis

- Extensive experience developing and using TradingView Pine Script for financial analysis and indicator-based modeling.
- Worked with Fibonacci Bollinger Bands, Volume Flow Indicator (VFI) and Wave Trend with Crosses.
- Used technical indicators and chart-based analysis to examine financial-market behavior and potential risk conditions.

Independent Security Researcher

Self-employed · Independent research

- Conducts independent security research following a vehicle incident involving a sabotaged and totaled car.
- Develops software and security models focused on authorization boundaries, system integrity, provenance and accountability.
- Investigates security concerns through architectural analysis, threat modeling, code review and the development of defensive mechanisms.
- Applies systems-safety principles to the design of security-conscious software and human–machine interactions.

Research and implementation: GitHub repository

Professional Driver — Rideshare, Delivery & Courier

Self-employed · Uber and Lyft platforms; independent courier and delivery work

- Provided passenger transportation through Uber and Lyft.
- Performed delivery and courier driving, alongside independent vehicle testing.
- Reports a recorded downhill speedometer reading exceeding **202 MPH**, with Porsche Stability Management disabled.
- Reports a driving run exceeding **14 miles at an approximately 200 MPH average**, alongside additional high-speed driving test runs.
- Has experience with multiple high-speed driving test runs and reports operating under conditions involving disabled vehicle stability systems.
- No longer conducts vehicle testing following the loss of access to a suitable vehicle.

High-speed figures are personal reports, not independently verified professional performance records.

Independent Wholesale Distributor & Import/Export Operator

Self-employed · Independent commercial activity

- Experience with import and export activities, including travel to conventions and inventory acquisition.
- Coordinated product ordering, inventory receipt and distribution.
- Shipped wholesale inventory to retailers and conducted direct sales to individual customers.
- Handled activities spanning inventory procurement, logistics, wholesale distribution and retail transactions.

EDUCATION

Arizona State University

Aerospace Engineering / Software Engineering: Attended · No degree completed

Peggy Payne Academy: High School · Graduated.

SELECTED RESEARCH, INTELLECTUAL PROPERTY & TECHNICAL WORK

- **HarshalOS** — Independent software and systems architecture project.
- **AGS++** — Accountability-Gated Systems, Accessories and Examples; original framework and technical documentation.
- **Keyed Human–Machine Hybrid Systems** — Research on structural accountability, bounded autonomy and human–machine responsibility.
- **Harshal's Systems** — Technical manuscript and associated intellectual property.

TECHNICAL COMPETENCIES

Area	Skills and experience
AI governance	Accountability architecture, bounded autonomy, human oversight
Systems safety	Refusal logic, state transitions, failure handling, reversibility
Software	JavaScript, browser-based systems, software architecture
Security	Threat modeling, security research, authorization and provenance
Financial analysis	TradingView, Pine Script, technical indicators, risk modeling
Commercial operations	Import/export, procurement, logistics, wholesale and direct sales
Transportation	Rideshare, delivery, courier operations and reported high-speed vehicle testing